A strict compound-channel minimax gap

Disproof of Conjecture 00000009542

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Abstract

Consider three genuine quantum channels from a qutrit to a classical bit: channel i reports whether the input basis index equals i. Each individually has maximized quantum mutual information $\log 2$. Their common-input max–min is instead $h(1/3) < \log 2$, where h is binary entropy. The compound entanglement-assisted coding theorem identifies the operational capacity with this common-input quantity, not the min–max asserted in the source. The calculation holds for every density matrix, including coherent inputs.

Three actual completely positive trace-preserving maps

Use input $A = \mathbb{C}^3$ with basis indexed by $0, 1, 2$, output $B = \mathbb{C}^2$, and environment $E = \mathbb{C}^3$. For $i \in \{0, 1, 2\}$ define $b_i(j) = 0$ if $j = i$, and $b_i(j) = 1$ otherwise. Set

$$K_{ij} = |b_i(j)\rangle\langle j|, \qquad \Phi_i(X) = \sum_{j=0}^{2} K_{ij} X K_{ij}^*.$$

This is a complex-linear map. Every matrix amplification is a sum of positive conjugations by $I \otimes K_{ij}$, so it is completely positive. Moreover $\sum_j K_{ij}^* K_{ij} = I_3$, proving trace preservation. Directly,

$$\Phi_i(X) = \operatorname{diag}(X_{ii}, \operatorname{tr} X - X_{ii}).$$

For a density matrix ρ, write $p_i = \rho_{ii}$. Positivity makes these real and nonnegative, and $p_0 + p_1 + p_2 = 1$. The output is the actual classical-bit state $\operatorname{diag}(p_i, 1 - p_i)$.

The genuine Stinespring isometry

$$V_i|j\rangle = |b_i(j)\rangle \otimes |j\rangle$$

satisfies $V_i^* V_i = I_3$. Tracing its output over E gives Φ_i. Tracing over B gives its complementary channel

$$\Phi_i^c(\rho) = P_i \rho P_i + Q_i \rho Q_i, \quad P_i = |i\rangle\langle i|, \quad Q_i = I_3 - P_i.$$

This is a two-block pinching, not full dephasing. Both output and complementary matrices are positive and have trace one.

Entropy and coherent inputs

Use natural logarithms, so capacities are in nats. Define $\eta(x) = -x \log x$ for $x > 0$ and $\eta(0) = 0$, $S(\rho) = \operatorname{tr} \eta(\rho)$, and

$$h(p) = \eta(p) + \eta(1-p).$$

The usual channel mutual information is

$$I(\rho, \Phi_i) = S(\rho) + S(\Phi_i(\rho)) - S(\Phi_i^c(\rho)).$$

To justify this standard entropy-exchange expression, purify ρ on a reference system R and apply V_i. The resulting state on RBE is pure, so its reduced states on RB and E have the same nonzero eigenvalues by Schmidt decomposition. Substituting $S(RB) = S(E)$ into the usual reference-output mutual information gives the displayed formula.

Here is a direct proof of the needed pinching entropy inequality, including coherent matrices. Let ρ have orthonormal eigenbasis u_b and eigenvalues λ_b, and let $\sigma = P\rho P + Q\rho Q$ have eigenbasis v_a and eigenvalues μ_a, where P, Q are complementary orthogonal projections. Set

$$w_{ab} = |\langle v_a, Pu_b \rangle|^2 + |\langle v_a, Qu_b \rangle|^2.$$

These weights are nonnegative. Completeness of both orthonormal bases and $P^2 + Q^2 = I$ give

$$\sum_b w_{ab} = 1, \qquad \sum_a w_{ab} = 1, \qquad \mu_a = \sum_b w_{ab} \lambda_b.$$

Since η is concave on $[0, \infty)$, finite Jensen and the column sums imply

$$S(\sigma) = \sum_a \eta(\mu_a) \geq \sum_{a,b} w_{ab} \eta(\lambda_b) = \sum_b \eta(\lambda_b) = S(\rho).$$

Applying this to P_i, Q_i yields, for every actual density matrix,

$$I(\rho, \Phi_i) \leq S(\Phi_i(\rho)) = h(p_i).$$

No restriction to diagonal or classical inputs was used in this upper bound. If ρ is diagonal, the pinching leaves it unchanged, and equality holds.

Exact max–min and min–max values

Binary entropy is at most $\log 2$, with equality at $p = 1/2$. Thus for every i and every state, $I(\rho, \Phi_i) \leq \log 2$. The diagonal state with $p_i = 1/2$ and the other two probabilities $1/4$ attains equality. Consequently

$$\sup_\rho I(\rho, \Phi_i) = \log 2 \quad \text{for each } i, \qquad \min_i \sup_\rho I(\rho, \Phi_i) = \log 2.$$

On the other hand, every state has some $p_i \leq 1/3$, because its three diagonal probabilities sum to one. Binary entropy is increasing on $[0, 1/2]$, so that channel satisfies

$$I(\rho, \Phi_i) \leq h(p_i) \leq h(1/3).$$

This proves $\sup_\rho \min_i I(\rho, \Phi_i) \leq h(1/3)$. The uniform diagonal state $\rho = I_3/3$ attains $I(\rho, \Phi_i) = h(1/3)$ simultaneously for all three channels. Hence

$$\sup_\rho \min_i I(\rho, \Phi_i) = h(1/3) < \log 2 = \min_i \sup_\rho I(\rho, \Phi_i).$$

The strict inequality is exact, not numerical: $h(1/3) = \log 3 - \frac{2}{3}\log 2$, so its gap from $\log 2$ is $\frac{1}{3}\log(32/27) > 0$.

The operational capacity bridge

For a compound memoryless quantum channel set $\mathcal{W}$, the channel is fixed during transmission but unknown within $\mathcal{W}$, and one code must be reliable for every member. The entanglement-assisted compound-channel coding theorem states

$$C_{\mathrm{EA}}(\mathcal{W}) = \sup_\rho \inf_{\Phi \in \mathcal{W}} I(\rho, \Phi).$$

This holds under both average and maximal transmission error; see [2, Theorem 1, equation (7)]. The reference uses bits; multiplying by $\log 2$ gives the same formula in the nats used here.

For our finite set $\mathcal{W} = \{\Phi_0, \Phi_1, \Phi_2\}$, the preceding exact calculation therefore gives

$$C_{\mathrm{EA}}(\mathcal{W}) = h(1/3),$$

whereas the source's proposed formula gives $\log 2$. The two cannot be equal. The relevant difference is the common encoder input needed for the unknown channel; each individual maximizing state need not work simultaneously for the whole set.

Conclusion and formalization scope

Theorem 1. *The entanglement-assisted capacity of a finite compound quantum channel set need not equal* $\min_i \max_\rho I(\rho, \Phi_i)$.

Proof. The three explicitly constructed CPTP maps satisfy $\sup_\rho \min_i I = h(1/3) < \log 2 = \min_i \sup_\rho I$. The compound coding theorem identifies the first quantity with their operational capacity. □

The first finite-set capacity formula is already false. No claim about the separate continuity statement or the additional error-exponent assertion is needed to refute the source's conjunction.

The Lean project uses genuine complex matrices with actual positive-semidefinite and trace-one conditions. Entropy is the real trace of the actual continuous functional calculus of $-x \log x$. Its spectral eigenvalue formula is proved, and the pinching entropy inequality is derived through the actual unitary eigenbases and finite Jensen argument above. No quantum entropy inequality is assumed as a custom axiom.

The channel is an actual complex-linear Kraus map. Complete positivity is proved for every finite matrix amplification, and trace preservation holds on every input matrix. The explicit Stinespring matrix is proved an isometry, and both actual partial-trace identities are verified. Positive normalized output and environment states are proved. Thus the mutual-information expression uses genuine state and channel entropies, not a chosen binary-entropy proxy. Its bound for all coherent states, diagonal equality cases and exact optimizing states are formal theorems.

The common-input optimization is an actual supremum over all density matrices of their finite infimum over the three channels. Individual optimizations are actual suprema over all states. Their exact values and strict gap are proved using global bounds and attaining states; conditional-supremum defaults do not supply the result.

The formal project verifies this actual single-letter gap. The operational coding theorem and the standard purification justification of the entropy-exchange formula are explicitly invoked in the mathematical proof, not newly formalized in Lean. In particular, the project does not define its optimization to be an operational capacity by fiat.

Reproducibility and references

Lean 4.19.0 and Mathlib revision
`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The complete project is rebuilt in a fresh directory. Printed theorem dependencies contain only standard foundational axioms. There are no proof placeholders, custom axioms or external computation oracles, and no auxiliary numerical verifier is needed. Compile this paper with `tectonic main.tex`.

The exact bilingual conjecture, source provenance, actual verification logs and artifact hashes are included. The author agent and parent performed separate internal reviews; no external independent review is claimed.

The cited coding theorem supplies the operational bridge. All model-specific channel constructions, entropy estimates and optimizations used in this counterexample are supplied explicitly above and checked in the accompanying project. The reference's Remark 3 also discusses a strict minimax gap for a different compound-channel example; our smaller three-channel construction is proved here.

References

[1] The Last Math Competition, Conjecture 00000009542. Exact bilingual source included as `SOURCE.md`.

[2] H. Boche, G. Janßen and S. Kaltenstadler, *Entanglement-assisted classical capacities of compound and arbitrarily varying quantum channels*, arXiv:1606.09314v3, 2017. Paper and author PDF. Theorem 1, equation (7), gives the compound capacity formula; equation (2) defines channel mutual information.